TypeScript has become a cornerstone of modern frontend and full-stack development, thanks to its powerful static typing and ability to catch errors early in the development process. By adding a type system on top of JavaScript, TypeScript improves code safety, readability, and maintainability —making it easier to build complex, scalable applications with confidence.

As more teams adopt TypeScript across UI and backend projects, its advanced tooling—such as intelligent autocompletion, refactoring support, and real-time type checking—has become indispensable for professional developers. In technical interviews, TypeScript proficiency is now often treated as a baseline skill, reflecting its critical role in modern web and enterprise development.

Question: "Why are more companies adopting TypeScript?"

Analysis: This question is designed to assess a candidate's understanding of TypeScript's value in modern software development and their awareness of industry trends. A strong answer demonstrates not only familiarity with TypeScript's technical features—such as static typing, advanced tooling, and integration with popular libraries—but also insight into how these features translate into business advantages.

Answer: More companies are adopting TypeScript because it significantly enhances code reliability, scalability, and maintainability in modern web development. Its static typing catches errors at compile time, reducing runtime bugs and lowering debugging costs. TypeScript's advanced tooling —such as autocompletion, intelligent navigation, and refactoring support— boosts developer productivity and code accuracy. Additionally, static types serve as self-documentation, improving collaboration across teams, especially in large or distributed environments. The growing adoption trend of TypeScript is illustrated in figure 3.4 (https://npmtrends.com/typescript).

Figure 3.4 Weekly npm downloads of TypeScript as of June 1, 2025: 88.5 million